

Functional Test - Module 9: Wait List

Project: Blue Team — Moffat Bay Marina

Course: CSD 460 - Capstone Project

Description: Test plan for the Wait List page (Sara — Front End, Miguel — Back End): the public per-slip-size line counts and wait estimates, the signed-in customer's own place in line, and leaving the wait list.

Date: 2026/09/23

Front End Developer: Sara White

Back End Developer: Miguel Fernandez

Front End Developer Tester: Sara White

Back End Developer Tester: Miguel Fernandez

Front End Peer Tester: Robert Breutzmann

Back End Peer Tester: Carolina Rodriguez

To complete the test case plan, fill out the information for Project, Course, Description and Date in the header above.

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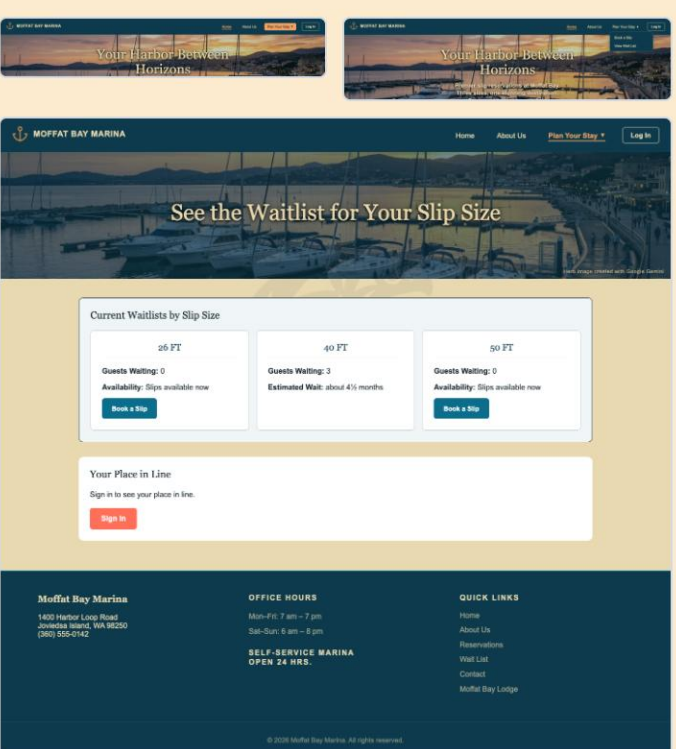
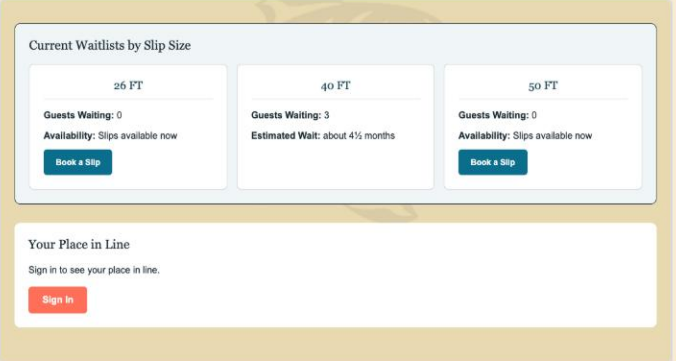
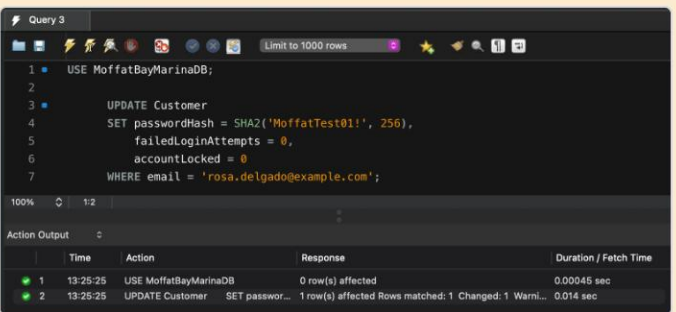
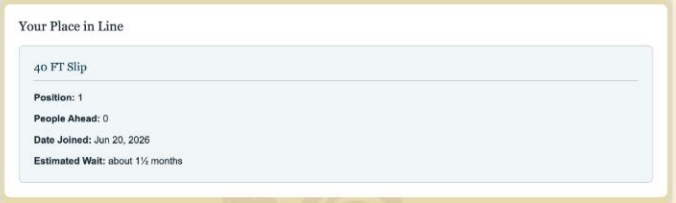
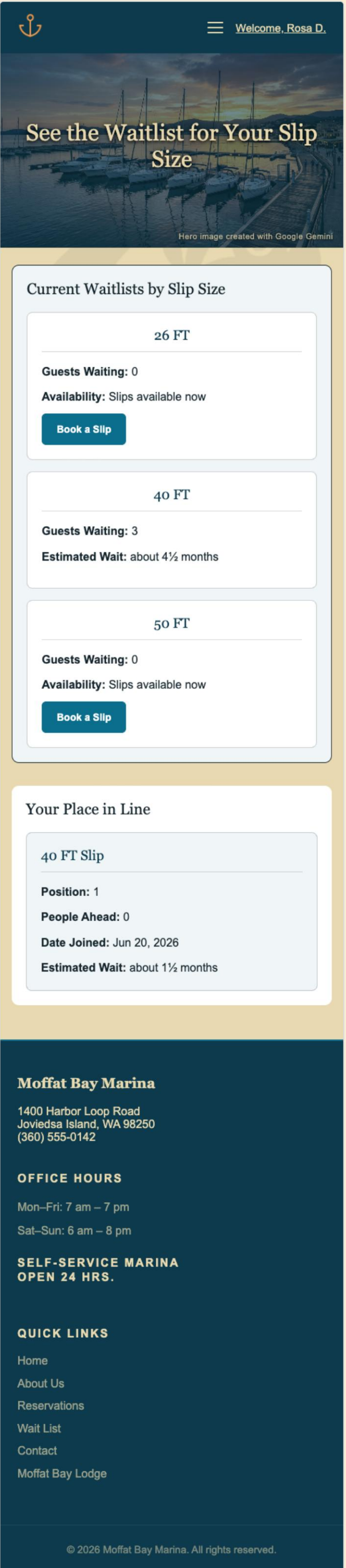
For each test, the developer should: provide a test description, a test objective, the developer name and date tested. For each step, fill out the actions to be taken and describe the expected results, and check Pass or Fail. The peer tester should provide their name, date tested, check Pass or Fail for each step, and add each step's screenshot(s) in the same cell as their Pass or Fail.

Test 1: Wait List Front-End Functionality

Test Objective: Verify that the Wait List page loads and displays correctly, navigation interactions function as expected, waitlist information is presented appropriately for logged-out and logged-in users and the page remains usable at a reduced browser width.

DEVELOPER: Sara White | TESTED ON: Sept 27, 2026

PEER TESTER: Robert Breutzmann | TESTED ON: Sept 27, 2026

Step	Action	Expected Results	Dev Test	Peer Test
1	Navigate to marinawebsite/index.jsp while logged out. Using the Plan Your Stay dropdown menu in the header, navigate to View Wait List.	The Plan Your Stay dropdown should highlight with the Golden Hour orange/gold color when hovered over. When selected, the dropdown menu should expand and the arrow should rotate. After selecting View Wait List, the page should load with the shared header, hero and footer sections. The waitlist content should include a card section titled Current Waitlists by Slip Size.	Pass	
2	Review the content displayed on the Waitlist page.	Cards should display for the available slip sizes. Each card should show the slip size, number of guests waiting and either an estimated wait or current availability. A separate card titled Your Place in Line should appear below the current waitlist information. While logged out, this section should display a Sign In button for the user to access their personal waitlist information.	Pass	
3	Run the SQL script provided below to reset the password for a customer who currently has an active waitlist entry.	<pre>USE MoffatBayMarinaDB; UPDATE Customer SET passwordHash = SHA2('MoffatTest01!', 256), failedLoginAttempts = 0, accountLocked = 0 WHERE email = 'rosa.delgado@example.com';</pre>	Pass	
4	Click the Sign In button in the Your Place in Line section and sign in using: Username: rosa.delgado@example.com Password: MoffatTest01!	After a successful login, the Your Place in Line section should update with the customer's waitlist information for a 40 ft slip. The information should display: - Position: 1 - People Ahead: 0 - Date Joined: Jun 20, 2026 - Estimated Wait: about 1½ months	Pass	
5	Resize the browser window to approximately one-quarter of its full-screen width.	The waitlist cards should stack appropriately for the reduced browser width. Card content should remain legible and clear, with no overlapping content or horizontal overflow.	Pass	

Comments

Developer (Sara)
Everything displayed correctly. I updated the **Date Joined** display to use the formatting getter provided in the JavaBean, making the date more legible for the user.

Peer Tester (Robert)
I had no issues while running this test. A couple of notes unrelated to the test that might want to be looked at.

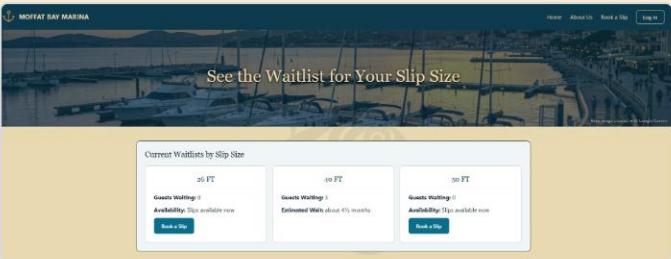
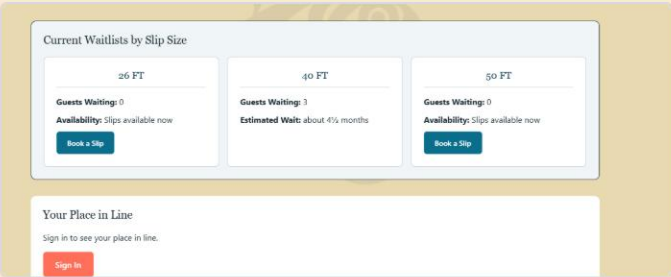
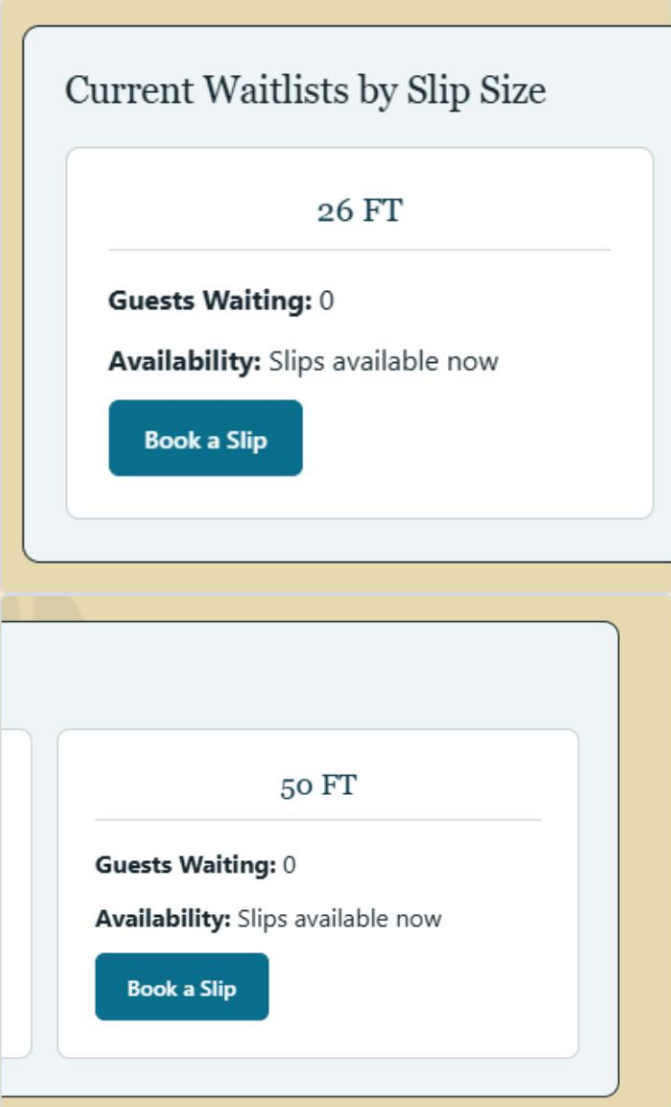
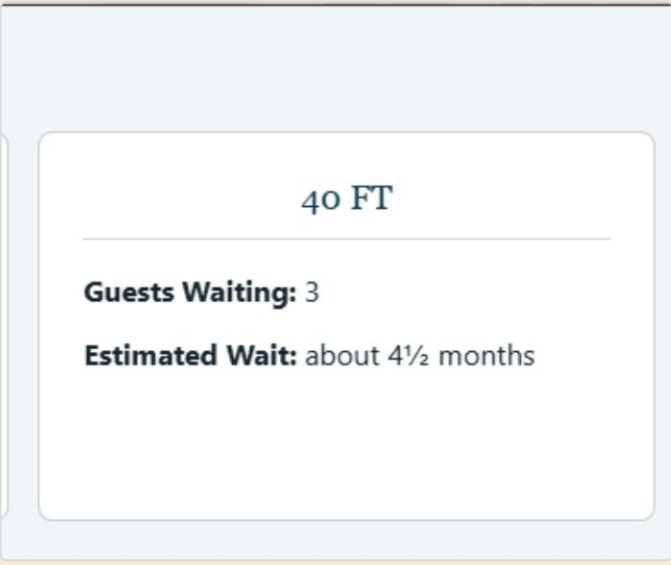
- The card for the current wait list display is quite wide and leaves a lot of white space. This could be looked at narrowing
- The password is reset during the test, but later used in Miguel's test as is. You should reset the password back to **MoffatDelgado01!** or reset the database.

Test 2: Public Wait List Figures Are Correct and Need No Sign In

Test Objective: Verify that `/waitList` serves the per-slip-size line counts and wait estimates to a signed-out visitor, that the counts match the database, and that the estimate counts only slips that are actually in service.

DEVELOPER: Miguel Fernandez | **TESTED ON:** Sept 26, 2026

PEER TESTER: Carolina Rodriguez | **TESTED ON:** Sept 26, 2026

Step	Action	Expected Results	Dev Test	Peer Test
1	In a private/incognito window, go to <code>http://localhost:8081/marina/website/waitList</code> without signing in.	The page loads normally - no sign-in panel, no redirect, HTTP 200. The header shows the Log In button rather than a welcome greeting.	Pass	Pass The full Wait List page loaded signed out, with the header showing Log In. 
2	Read the three slip size cards under "Current Waitlists by Slip Size".	All three sizes are shown - 26, 40 and 50 - even though only the 40 ft list has anyone on it. A size with nobody waiting still gets a card reading 0.	Pass	Pass The three cards together, showing 26 ft and 50 ft at 0 and 40 ft at 3. 
3	Read the 26 ft and 50 ft cards.	Both read <code>Guests Waiting: 0</code> and <code>Availability: Slips available now</code> with a Book a Slip link, and show no wait estimate - a size with free slips is offered, not estimated.	Pass	Pass The 26 ft card, and 50 ft cards, showing 0 waiting with the Book a Slip link and no estimate. 
4	Read the 40 ft card.	<code>Guests Waiting: 3</code> and <code>Estimated Wait: about 4 1/2 months</code> . No Book a Slip link, because every 40 ft slip is taken.	Pass	Pass The 40 ft card, showing 3 waiting and the estimate. 
5	Run the first query below against the database.	Exactly 3 rows come back, all at <code>sizeFt 40</code> , all <code>Waiting</code> - matching the count on the card. Desmond Okafor's <code>Fulfilled</code> row is not among them.	Pass	Pass The query result showing the three Waiting rows at 40 ft. 
6	Run the second query below, which counts slips by size and status.	40 ft has 23 operational slips and 1 in maintenance, not 24. The estimate uses 23: a slip under maintenance never comes free, so it cannot shorten anyone's wait. That is what makes the card read 4 1/2 rather than 4 - 4 x 24 ÷ 23 = 4.17, rounded up to the next half month.	Pass	Pass The query result showing 23 operational and 1 maintenance at 40 ft. 

For step 5, the three `Waiting` entries the 40 ft card is counting:

```
USE MoffatBayMarinaDB;

SELECT w.waitListID, w.customerID, c.email, sz.sizeFt, w.status, w.timeJoined
FROM WaitList w
JOIN SlipSize sz ON sz.slipSizeID = w.slipSizeID
JOIN Customer c ON c.customerID = w.customerID
WHERE w.status IN ('Waiting', 'Offered')
ORDER BY sz.sizeFt, w.timeJoined;
```

For step 6, the slip counts the estimate's N comes from:

```
USE MoffatBayMarinaDB;

SELECT sz.sizeFt, s.slipStatus, COUNT(*) AS slips
FROM Slip s
JOIN SlipSize sz ON sz.slipSizeID = s.slipSizeID
GROUP BY sz.sizeFt, s.slipStatus
ORDER BY sz.sizeFt;
```

Comments

Developer (Miguel)
All six steps passed. This page is deliberately the one page on the site that does not require a session - someone deciding whether to keep a boat here should be able to read how long the line is before they have an account - so the servlet treats a missing `customerId` as normal rather than as an error. Nothing about a particular person is public: the cards are counts only, no names, no join dates.
Step 6 is the one worth reading twice. I expected the 40 ft card to say "about 4 months" and it said 4 1/2, which turned out to be right: one 40 ft slip is in maintenance in the seed data, and a slip that is out of service never turns over, so it is correctly left out of the estimate. The seed data happens to exercise that rule on all three sizes.

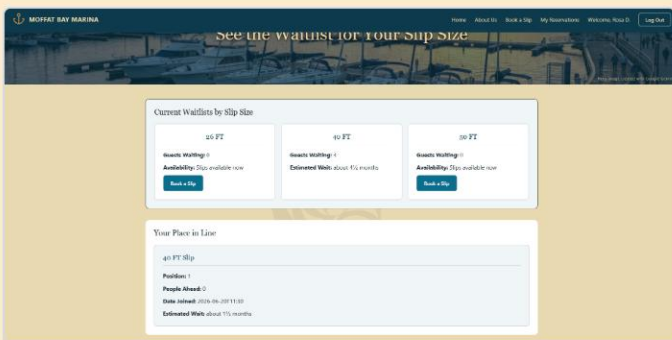
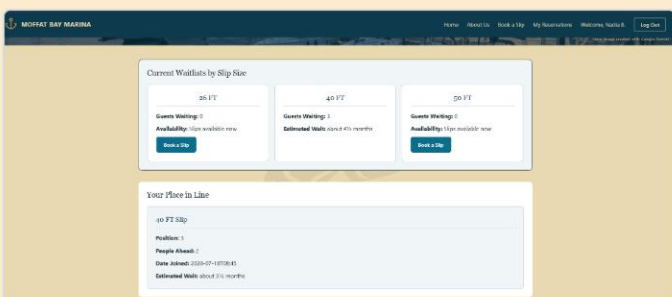
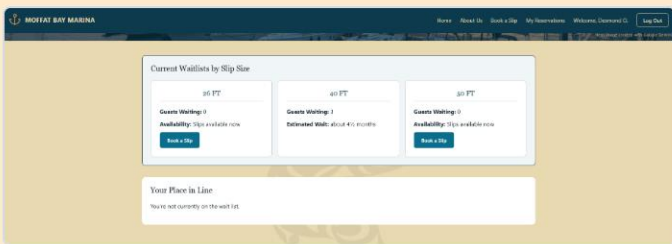
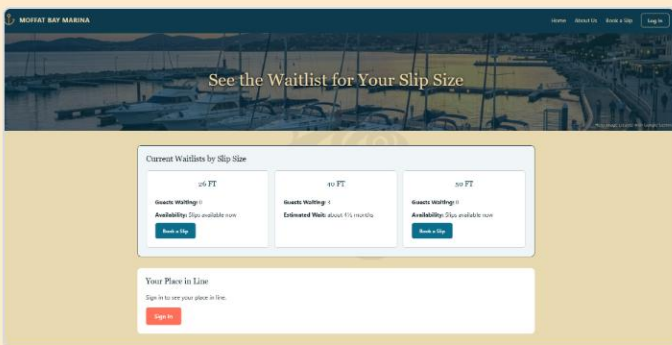
Peer Tester (Carolina)
The Wait List worked as expected while signed out. I confirmed that all three slip sizes were shown, including the sizes with 0 guests waiting. The 40 ft count matched the database results, and the wait estimate used the 23 operational slips rather than including the slip in maintenance. The 26 ft and 50 ft cards correctly showed that slips were available now instead of displaying a wait estimate.

Test 3: A Signed-In Customer Sees Only Their Own Place in Line

Test Objective: Verify that the "Your Place in Line" section is built from the session customer alone, that position and people-ahead are worked out from `timeJoined` (BR-20), and that a closed entry never appears.

DEVELOPER: Miguel Fernandez | TESTED ON: Sept 26, 2026

PEER TESTER: Carolina Rodriguez | TESTED ON: Sept 26, 2026

Step	Action	Expected Results	Dev Test	Peer Test																
1	Sign in as <code>rosa.delgado@example.com</code> / <code>MoffatDelgado04!</code> and open <code>/waitList</code> .	A "Your Place in Line" section appears below the public cards: 40 FT Slip, Position: 1, People Ahead: 0, and Estimated Wait: about 1½ months. She joined first, on 2026-06-20.	Pass	Pass <i>Rosa's entry card, showing position 1 with 0 people ahead.</i> 																
2	Sign out, then sign in as <code>nadia.brennan@example.com</code> / <code>MoffatBrennan06!</code> and open <code>/waitList</code> .	Position: 3, People Ahead: 2, Estimated Wait: about 3½ months. She joined last, on 2026-07-18, so Rosa and Arthur Penhale are both ahead of her.	Pass	Pass <i>Nadia's entry card, showing position 3 with 2 people ahead.</i> 																
3	Compare the two positions against the third query below.	The order on screen matches <code>timeJoined</code> ascending: Rosa (Jun 20) is 1, Arthur (Jul 5) is 2, Nadia (Jul 18) is 3. Position is derived, not stored - there is no position column on <code>WaitList</code> .	Pass	Pass <i>The query result, with the three entries in timeJoined order.</i> <table><tr><th>email</th><th>status</th><th>timeJoined</th><th>peopleAhead</th></tr><tr><td><code>rosa.delgado@example.com</code></td><td>Waiting</td><td>2026-06-20 11:30:00</td><td>0</td></tr><tr><td><code>arthur.penhale@example.com</code></td><td>Waiting</td><td>2026-07-05 14:15:00</td><td>1</td></tr><tr><td><code>nadia.brennan@example.com</code></td><td>Waiting</td><td>2026-07-18 08:45:00</td><td>2</td></tr></table>	email	status	timeJoined	peopleAhead	<code>rosa.delgado@example.com</code>	Waiting	2026-06-20 11:30:00	0	<code>arthur.penhale@example.com</code>	Waiting	2026-07-05 14:15:00	1	<code>nadia.brennan@example.com</code>	Waiting	2026-07-18 08:45:00	2
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<code>nadia.brennan@example.com</code>	Waiting	2026-07-18 08:45:00	2																	
4	Sign out, then sign in as <code>desmond.okafor@example.com</code> / <code>Moffat0kafor02!</code> and open <code>/waitList</code> .	The section reads "You're not currently on the wait list." rather than showing a card. His only entry is <code>Fulfilled</code> - closed entries are never in line. The public 40 ft count still reads 3, unchanged by his closed row.	Pass	Pass <i>Desmond signed in, showing the not-on-the-list message with the 40 ft count still at 3.</i> 																
5	Log out from the header, then reload <code>/waitList</code> .	The page still loads. The personal section is replaced by the sign-in prompt, and the three public cards are unchanged from Test 2.	Pass	Pass <i>The page after logging out, back to the sign-in prompt with the public cards intact.</i> 																

For step 3, the line in the order the page should show it:

```
USE MoffatBayMarinaDB;

SELECT c.email, w.status, w.timeJoined,
  (SELECT COUNT(*)
   FROM WaitList a
   WHERE a.slipSizeID = w.slipSizeID
        AND a.status IN ('Waiting', 'Offered')
        AND (a.timeJoined < w.timeJoined
              OR (a.timeJoined = w.timeJoined AND a.waitListID < w.waitListID))
  ) AS peopleAhead
FROM WaitList w
JOIN SlipSize sz ON sz.slipSizeID = w.slipSizeID
JOIN Customer c ON c.customerID = w.customerID
WHERE sz.sizeFt = 40
      AND w.status IN ('Waiting', 'Offered')
ORDER BY w.timeJoined;
```

Comments

Developer (Miguel)
All five steps passed.
Two customers rather than one on purpose. Position 1 with 0 ahead is exactly what a broken query returns for everybody, so Rosa on her own proves nothing - Nadia at position 3 with 2 ahead is what actually shows the ordering works. Step 4 is the other half of it: a `Fulfilled` entry has to be invisible here and must not be counted in the public number either.
`peopleAhead` is counted in SQL rather than by loading the line into Java, so no other customer's row is ever read into memory to work out where you stand. The customer ID comes from the session and is never read off the request, so there is no parameter to edit to see someone else's place.

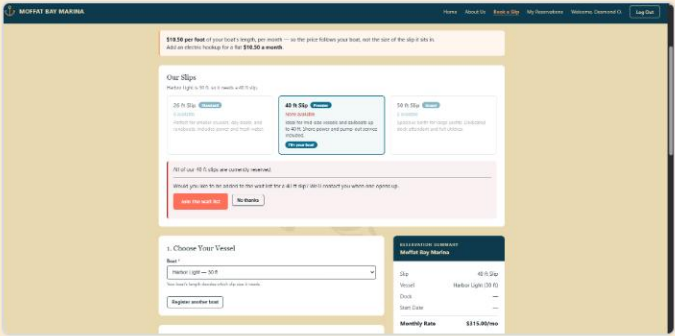
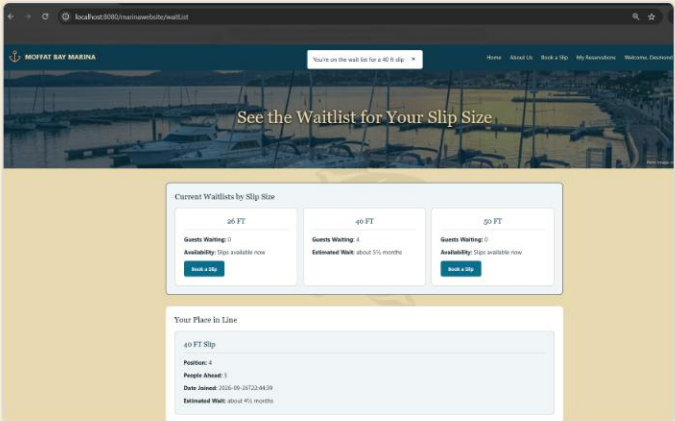
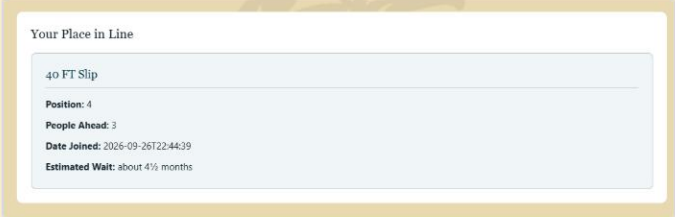
Peer Tester (Carolina)
The signed-in wait list information worked correctly for each customer I tested. Rosa showed as first in line with nobody ahead of her, while Nadia showed as third with two people ahead. The database query matched the same order based on the time each customer joined. Desmond did not show as currently waiting because his entry was already fulfilled. After signing out, the personal section went away but the public wait list information still stayed available.

Test 4: Joining From the Reservation Page Lands on the Servlet and Updates Both Sections

Test Objective: Verify that joining the wait list writes a `Waiting` row, redirects through `/waitList` rather than straight at the JSP, and that both the public count and the customer's own position reflect the new entry immediately.

DEVELOPER: Miguel Fernandez | **TESTED ON:** Sept 26, 2026

PEER TESTER: Carolina Rodriguez | **TESTED ON:** Sept 26, 2026

Step	Action	Expected Results	Dev Test	Peer Test
1	Signed in as <code>desmond.okafor@example.com</code> , go to Book a Slip and choose Harbor Light from the vessel dropdown.	Harbor Light is 30 ft, so the page matches it to a 40 ft slip and the 40 ft card reads "None available". The wait list prompt appears with a Join the wait list button.	Pass	Pass The Reservation page with Harbor Light selected and the wait list prompt showing. 
2	Click "Join the wait list", then read the browser address bar.	The browser lands on <code>/marinawebsite/waitList</code> - the servlet - not <code>waitListLookup.jsp</code> . The "You're on the wait list for a 40 ft slip" message appears.	Pass	Pass The address bar reading <code>/marinawebsite/waitList</code>, with the confirmation message showing. 
3	Read the 40 ft card on the page you just landed on.	<code>Guests Waiting: 4</code> and <code>Estimated Wait: about 5½ months</code> - both up from 3 and 4½ before joining.	Pass	Pass The 40 ft card after joining, reading 4 waiting. 
4	Read the "Your Place in Line" section on the same page.	<code>Position: 4</code> , <code>People Ahead: 3</code> , and <code>Estimated Wait: about 4½ months</code> . The customer's own estimate and the public one are different figures and both are correct: 5½ is what somebody joining now would wait, 4½ is what Desmond waits from position 4.	Pass	Pass Desmond's entry card, showing position 4 with 3 people ahead. 
5	Run the query below.	One new <code>Waiting</code> row for Desmond at 40 ft, with <code>timeJoined</code> set to the moment he joined and <code>timeClosed</code> null. His older <code>Fulfilled</code> row is untouched - joining again adds an entry rather than reopening a closed one.	Pass	Pass The query result showing Desmond's new Waiting row alongside his older Fulfilled one. 

For step 5, both of Desmond's wait list rows:

```
USE MoffatBayMarinaDB;

SELECT w.waitListID, sz.sizeFt, w.status, w.timeJoined, w.timeClosed
FROM WaitList w
JOIN SlipSize sz ON sz.slipSizeID = w.slipSizeID
JOIN Customer c ON c.customerID = w.customerID
WHERE c.email = 'desmond.okafor@example.com'
ORDER BY w.timeJoined;
```

Comments

Developer (Miguel)
All five steps passed, but only after a fix. `reservation.js` sent the customer to `waitListLookup.jsp` directly after joining. That was correct while the page was a Coming Soon stub, but once `WaitListServlet` existed it skipped `doGet()` , so the page rendered with no summaries at all and told someone who had just joined the wait list that they were not on it. Changed to `/waitList` . It is Robert's file and I flagged it in Discord and in the PR rather than changing it quietly, but my half does not work without it - it is the same class of bug as the login redirect I fixed in Module 8, and the site's convention is that every link goes through a servlet.
Step 4 is the step I would keep if I could only keep one. Two estimates from the same query disagreeing in the right direction is what shows the estimator is using each position rather than one number for the whole page.

Peer Tester (Carolina)
Joining the wait list from the Reservation page worked as expected. After selecting Harbor Light, I was able to join the 40 ft wait list and was sent to the `/waitList` page instead of directly to the JSP. The public wait list count increased right away, and Desmond's personal section showed his new position with the correct number of people ahead. I also checked the database and confirmed that a new `Waiting` row was added while his older `Fulfilled` entry stayed unchanged.